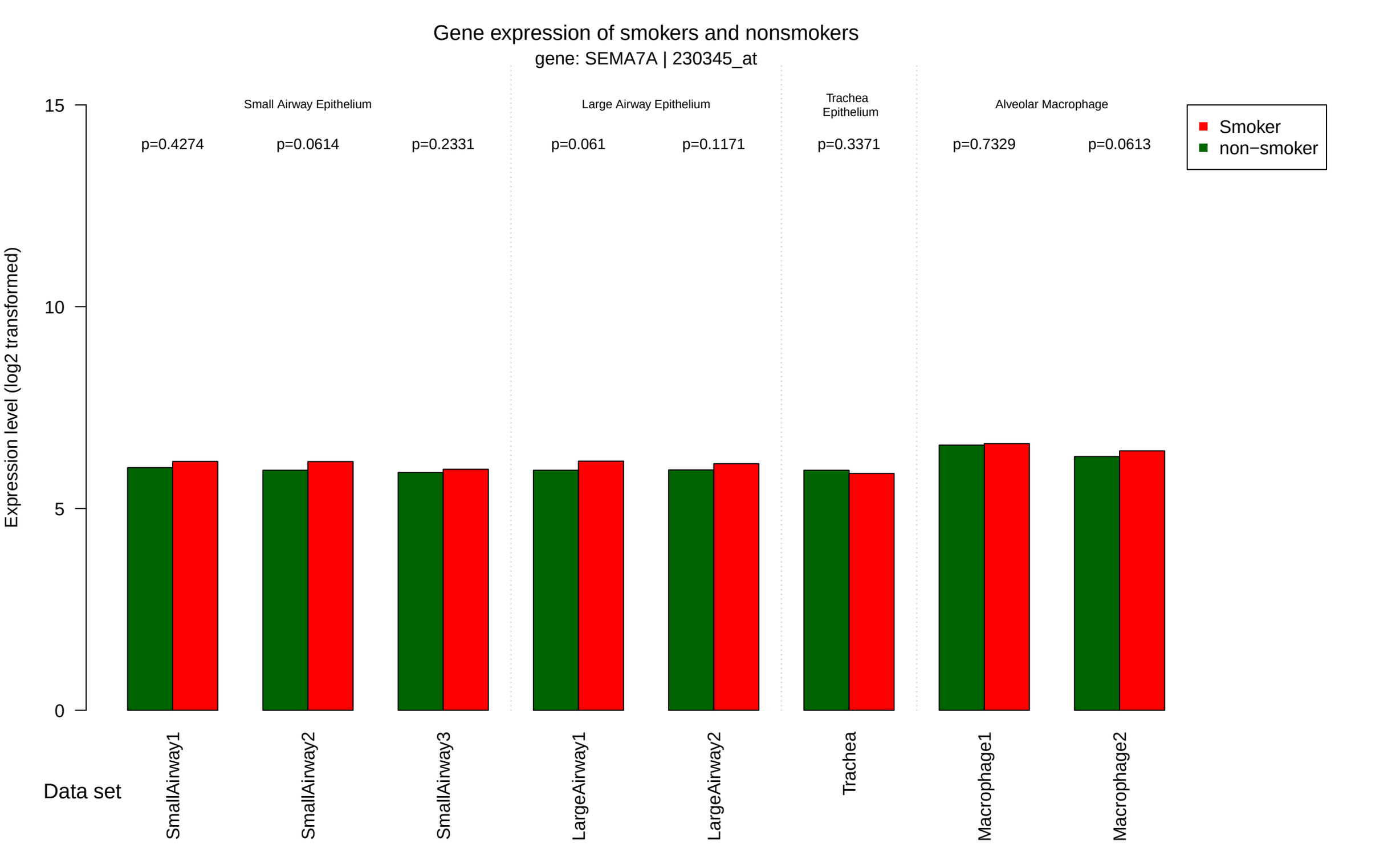
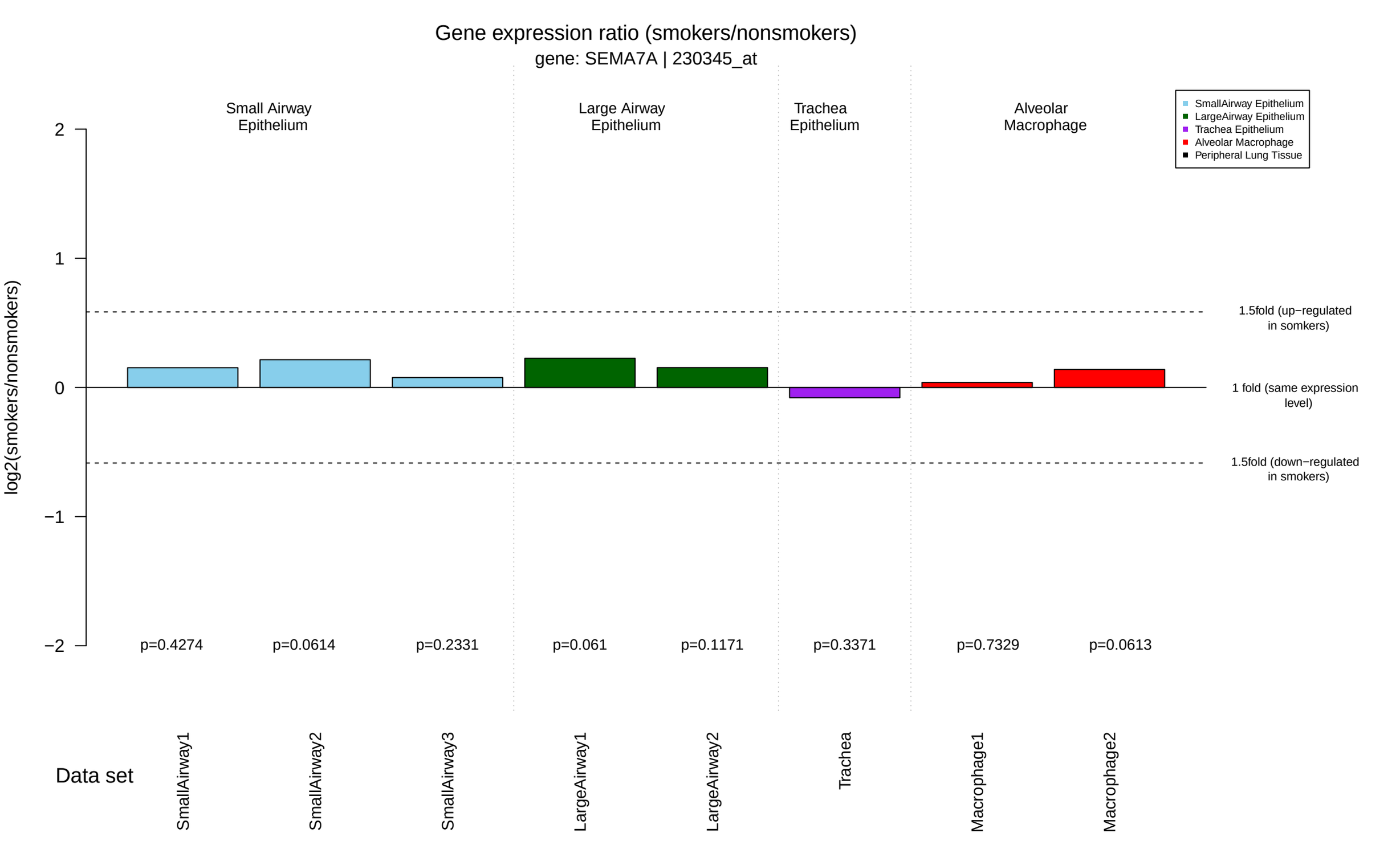
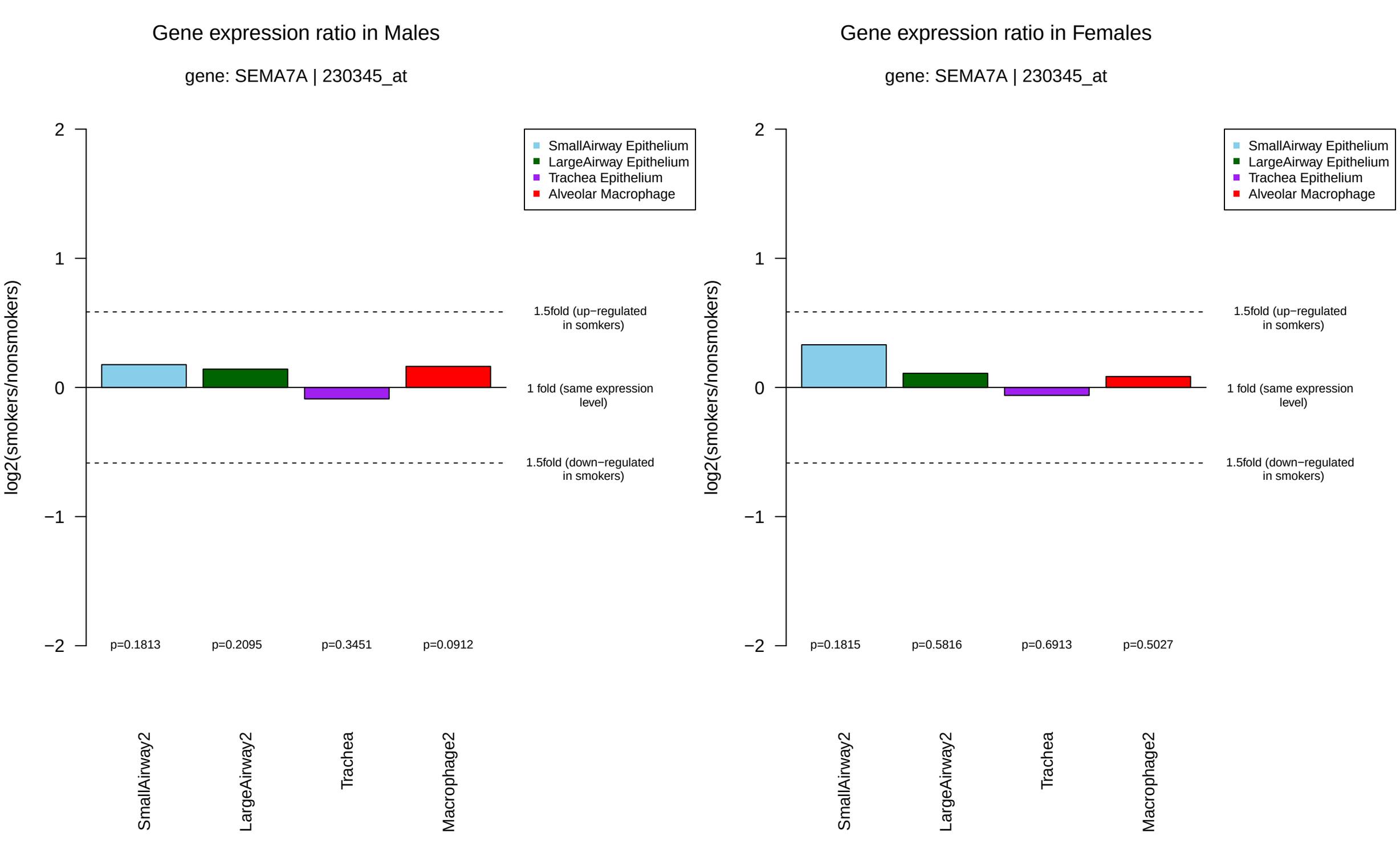
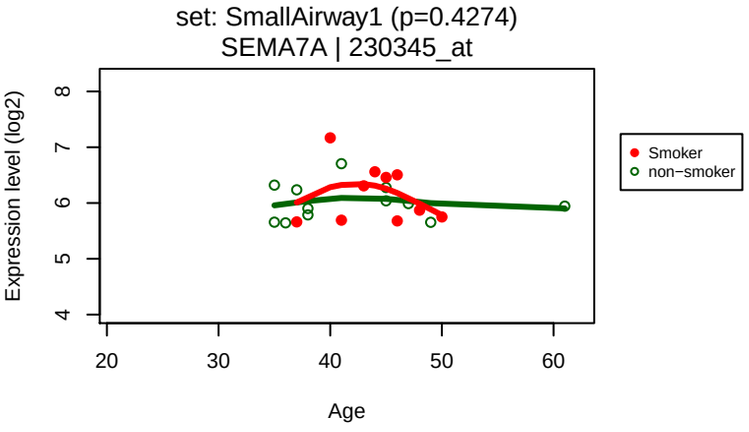
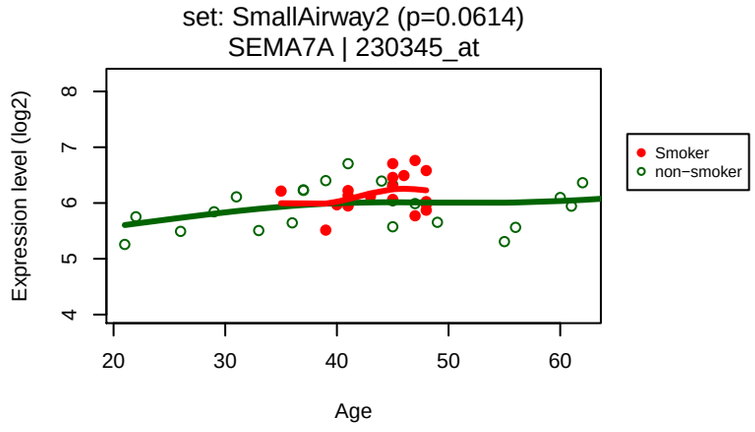
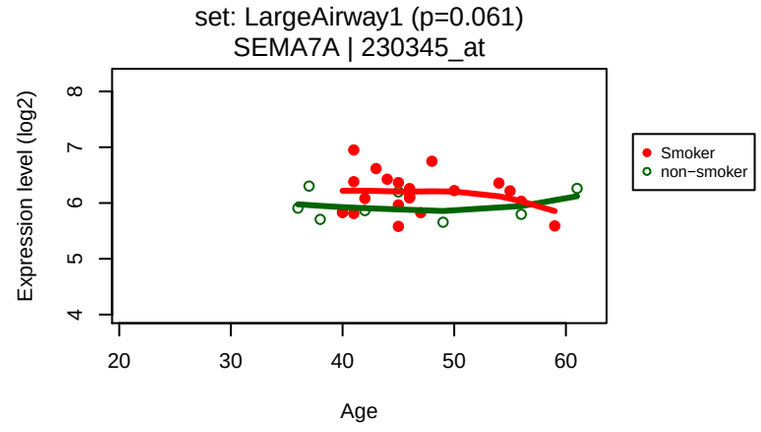

Gene expression of smokers and nonsmokers
gene: SEMA7A | 230345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4274
p=0.0614
p=0.2331
p=0.061
p=0.1171
p=0.3371
p=0.7329
p=0.0613
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

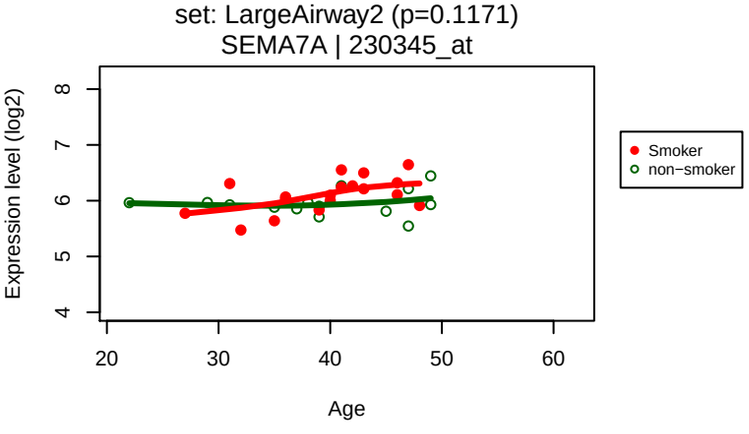

Gene expression ratio (smokers/nonsmokers)
gene: SEMA7A | 230345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4274
p=0.0614
p=0.2331
p=0.061
p=0.1171
p=0.3371
p=0.7329
p=0.0613
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

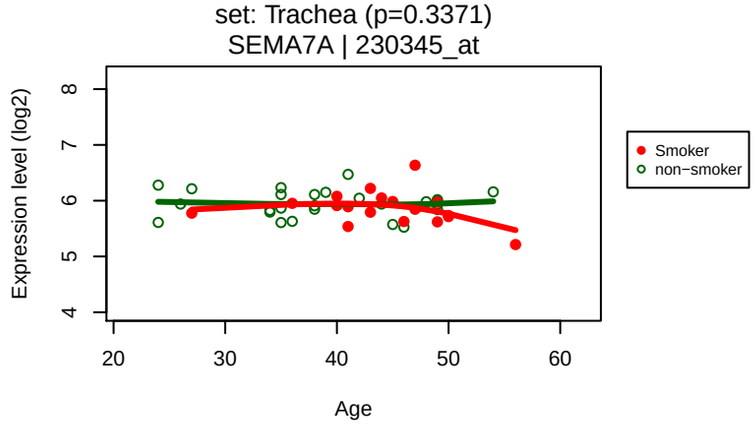

Gene expression ratio in Males
gene: SEMA7A | 230345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1813
p=0.2095
p=0.3451
p=0.0912
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA7A | 230345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1815
p=0.5816
p=0.6913
p=0.5027
SmallAirway2
LargeAirway2
Trachea
Macrophage2

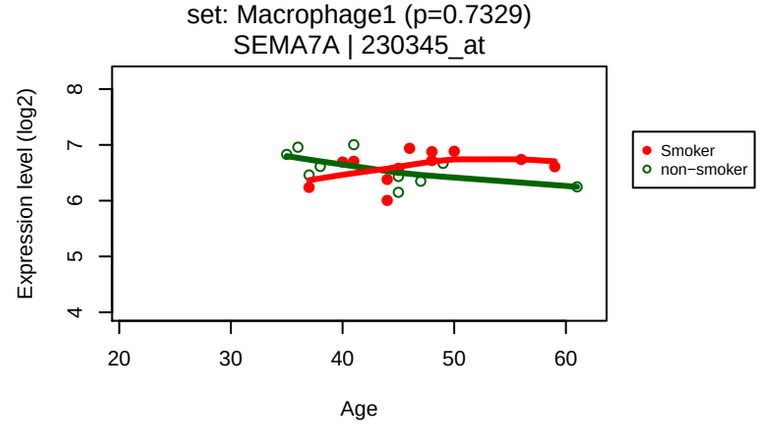

set: SmallAirway1 (p=0.4274)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker

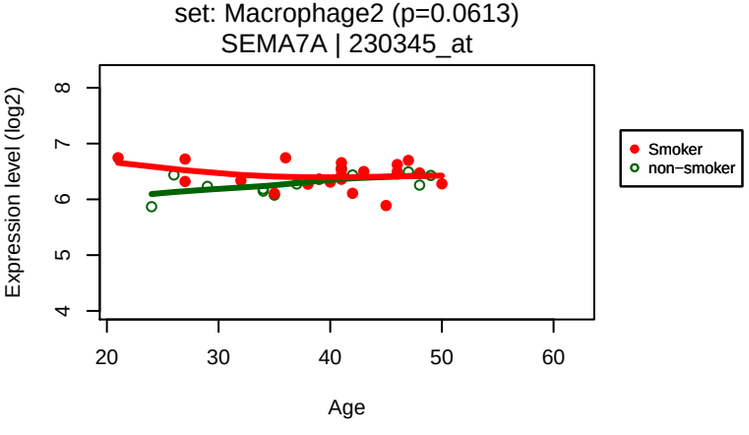

set: SmallAirway2 (p=0.0614)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker

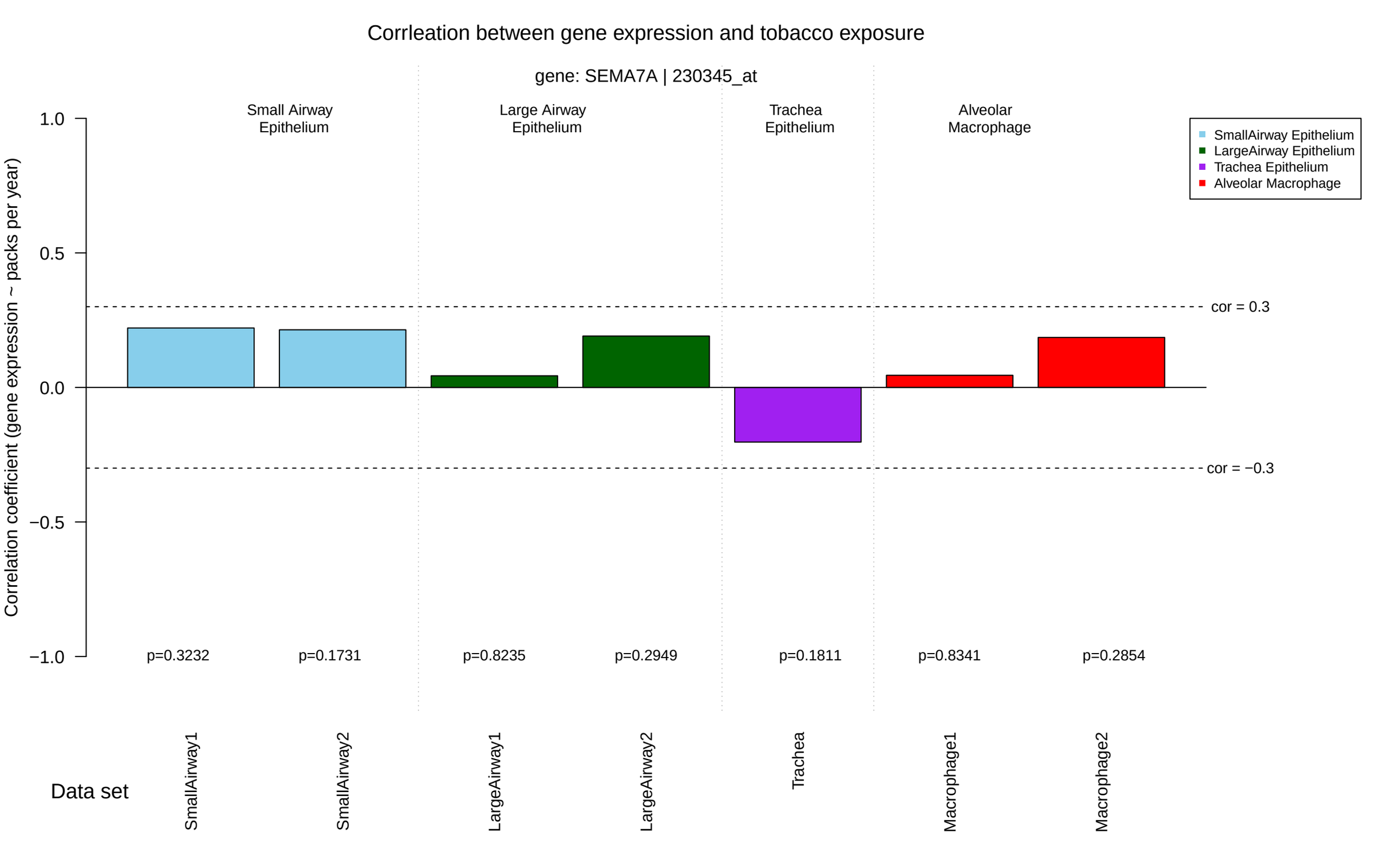

set: LargeAirway1 (p=0.061)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1171)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3371)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7329)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0613)
SEMA7A | 230345_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA7A | 230345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3232
p=0.1731
p=0.8235
p=0.2949
p=0.1811
p=0.8341
p=0.2854
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2